

Protocol, Not Architecture: What Seven Pipelines on One Predictive-Maintenance Benchmark Actually Measure

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Paper under double-blind review. Do not distribute.

Abstract

We report an audit of a public hydraulic-pump predictive-maintenance benchmark in which we built seven analysis pipelines (B–G) as separate work-streams over the same two CSV files, deliberately varying windowing, splitting, threshold sourcing, scoring unit and post-processing. Reported F_1 across these pipelines spans 0.684 to 0.988 — a range wider than most architecture comparisons in this literature — yet no pipeline is meaningfully better at detection: wherever we measure it under a leakage-controlled protocol, average precision is ≥ 0.98 and recall saturates at 1.000. We isolate the cause. Holding the network fixed and changing only the protocol moves the official LSTM-Autoencoder from $F_1 = 0.684$ to 0.988 in one pipeline and from 0.743 to 0.959 in another. We then show three consequences that any single-pipeline study would have missed. (i) Detection metrics cannot order models, but robustness to benign measurement shift separates them by more than $5\times$ (+17.0 vs. +89.7 percentage points of added false alarms). (ii) Cross-validated false-alarm rate does not predict deployed false-alarm rate: on the identical held-out block, five-fold CV reports 0.27% while the frozen deployment configuration produces 8.23%, a $30\times$ gap caused solely by which block supplies the conformal calibration set. (iii) An anomaly score that ranks perfectly can still be useless as a probability — one pipeline’s risk score attains Brier 0.487 against 0.111 for a constant predictor. Finally we show the benchmark’s label is confounded with acquisition session, and that observation length alone bounds detectability (recall $0.461 \rightarrow 0.821$ as windows grow from 4 to 30 samples). We argue that packaged reference analyses should report acquisition structure, a parameter-free baseline, and a measurement-shift stress test, and we explain why the seven F_1 values in this paper must not be ranked against each other.

1 Introduction

A public dataset shipped with a reference implementation is a common entry point into industrial anomaly detection, and the reference score becomes the number to beat. This paper asks what that number measures.

Our setting is a fine-blanking press hydraulic pump monitored by two vibration channels and one current channel, distributed with a guidebook implementing an LSTM-Autoencoder that reports $F_1 = 0.748$ (Ministry of SMEs and Startups and KAIST, 2022). We set out to beat it. Instead we built seven pipelines, found that essentially all of them saturate the detection metric, and found that the published number is reproducible but uninformative.

The seven pipelines (which we label B through G, following their internal development order) were constructed as separate work-streams with different design commitments: deep reconstruction models, isolation forests, PCA-based multivariate statistical process control, shrinkage Mahalanobis monitors, a one-line range rule, and supervised logistic regression. Crucially they also differ in *protocol*: window length and whether windows may cross acquisition gaps; whether splits are random, time-ordered, or blocked by acquisition burst; whether the decision threshold is fit on normal data alone or on a validation set containing fault labels; whether scores are evaluated per row, per window, or per burst; and whether alarms are post-processed.

Contributions.

(1) Protocol dominates architecture on this benchmark (§5). Holding the reference network, optimiser and epoch budget fixed and changing only preprocessing, windowing, score aggregation and threshold sourcing moves F_1 from 0.684 to 0.988 in one pipeline and from 0.743 to 0.959 in another. The 0.743 figure reproduces the published 0.748 to within seed noise, so the gap is not a reimplementing artefact.

(2) Detection saturates, so something else must do the discriminating (§6). Under a leakage-controlled protocol a parameter-free range rule reaches $F_1 = 0.973$, higher than every learned model we tested. Measurement-shift robustness, by contrast, spreads the same models across a $5\times$ range.

(3) Cross-validation does not predict deployment (§7). We exhibit a $30\times$ discrepancy on the identical evaluation block, attributable entirely to the choice of conformal calibration block — a direct, quantified failure of exchangeability under blocked splitting.

(4) Perfect ranking is compatible with useless probabilities (§8). A score with $AP \geq 0.99$ achieves a Brier score four times worse than predicting the base rate.

(5) A negative methodological result (§10). We explain

why the seven F_1 values reported here cannot be assembled into a ranking, and we decline to do so. We regard publishing the incomparability, rather than a leaderboard, as the correct output of this study.

We are explicit about what this study is not. The seven pipelines were built by one group, not by independent teams; this is a protocol-variation study, not a replication study. The benchmark contains exactly one fault event, recorded five days after the normal data, so separability is confounded with acquisition session and no lead-time claim is measurable.

2 Related Work

Evaluation pathologies in time-series anomaly detection. Kim et al. (2022) show that the widely used point-adjustment protocol inflates F_1 so severely that random scores appear state-of-the-art. Wu and Keogh (2023) argue that several standard benchmarks are trivially solvable or mislabelled, and Garg et al. (2022) reach similar conclusions systematically. Our finding is adjacent but distinct: we use no point adjustment and the labels are not wrong. The benchmark is simply so separable that the metric stops ordering methods, while protocol choices that are usually treated as implementation details move the headline number by more than any architecture choice does.

Shortcut learning and confounded acquisition. That models key on nuisance correlates rather than the phenomenon of interest is documented in vision and medical imaging (Geirhos et al., 2020; Zech et al., 2018), where site or scanner identity is easier to predict than pathology; Recht et al. (2019) show how fragile benchmark conclusions are to distribution details. Here class is perfectly aligned with acquisition date, which is the starkest possible form.

Classical multivariate process monitoring. Hotelling’s T^2 with squared prediction error is standard for industrial monitoring (Jackson and Mudholkar, 1979; Qin, 2003) and yields per-variable contributions for triage. We use it alongside shrinkage covariance estimation (Ledoit and Wolf, 2004) and Isolation Forest (Liu et al., 2008).

Distribution-free calibration. Several of our pipelines set thresholds by split-conformal p -values (Vovk et al., 2005; Angelopoulos and Bates, 2023). Overlapping windows violate exchangeability, so we block by acquisition burst in the spirit of purged cross-validation (López de Prado, 2018); §7 quantifies what happens when that is not enough.

3 The Benchmark

The package contains a normal record of 20,000 rows (2022-07-12) and a fault record of 600 rows (2022-07-17), each with upper and lower pump vibration (AI0, AI1), motor

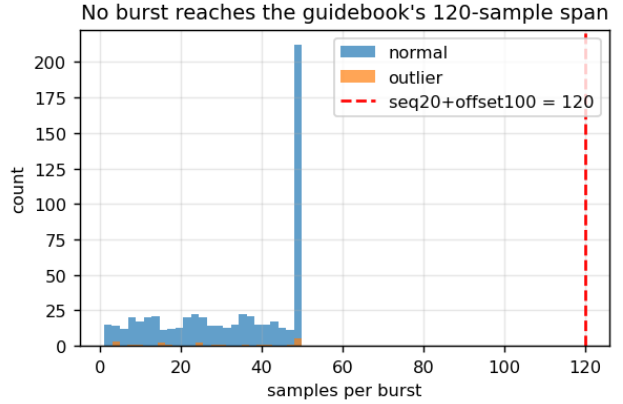


Figure 1: Burst length distribution. The reference analysis span (120 samples, dashed) exceeds every burst in both files.

current (AI2), and an equipment-state flag that is constant within each file. Sampling is documented as 10 Hz.

Acquisition is burst-sampled. The normal file spans 4615.8 s of wall-clock time while containing only 2000 s of samples at 10 Hz; 598 inter-sample gaps exceed 0.5 s. Segmenting there yields 599 bursts, median 37 and maximum 50 samples — at most 4.9 s of contiguous signal. The fault file has 21 bursts, maximum 50 samples. **No burst reaches the 120 samples that the reference pipeline’s 20-sample window plus 100-step label offset spans** (Figure 1), so that pipeline’s windows necessarily cross gaps and its stated 10 s horizon is unrealizable. Restricting windows to lie inside one burst removes 25.6% of windows at length 10 and 50.1% at length 20.

One fault event, confounded with date. The fault file is a single uninterrupted recording of an already-faulted machine. Its 428 windows (at length 10) are overlapping fragments of *one* event, not 428 independent positives, and no pre-failure segment exists. Normal and fault were recorded five days apart, so load, tooling, product and sensor state are free to differ. Every separability result below must be read against this.

A preprocessing step that removes the deep model’s advantage. The reference pipeline applies $|\cdot|$ to all three channels. Window RMS is *exactly* invariant to this ($|x|^2 = x^2$), as are mean absolute value and peak: measured over 14,871 normal windows the maximum change in each is 0.0. What the transform destroys is sign and phase — window standard deviation changes by up to 1.19×10^2 , signed mean by 2.78×10^2 , peak-to-peak by 3.04×10^2 and kurtosis by 4.88. The signed mean matters: the fault record’s current channel carries a large DC offset that $|\cdot|$ confounds with an amplitude increase. The reference thus discards

the only information a sequence model could exploit that a moment-based monitor cannot, before training the sequence model.

4 Seven Pipelines

Table 1 summarises how the pipelines differ. The differences are not incidental: each row changes what the reported number means.

Pipeline D additionally reproduces the reference LSTM-Autoencoder exactly ($64 \rightarrow 32 \rightarrow \text{repeat} \rightarrow 32 \rightarrow 64$) in two configurations, G0 following the guidebook and G1 inside D’s own protocol, which gives us a controlled architecture-held-fixed comparison. Pipeline B independently ablates the same network one factor at a time.

5 Protocol Dominates Architecture

Table 2 reports both controlled experiments. In pipeline B, changing only the score aggregation (last-timestep to whole-window MSE) raises F_1 from 0.684 to 0.826; fitting the threshold on normal data alone instead drives false positives to zero but recall to 0.537; combining all structural corrections yields $F_1 = 0.988 \pm 0.001$. In pipeline D, the guidebook configuration reproduces the published result (0.743 ± 0.029 vs. published 0.748; precision, recall and FPR all within seed noise), and the same network inside D’s protocol reaches 0.959.

Two independent routes therefore give the same answer: on this benchmark the evaluation protocol is the first-order variable and the architecture is not. We also note that under G0 early stopping never fired — all three seeds consumed the full 800-epoch budget despite a patience of 40 on training loss — whereas G1 stopped between 499 and 679 epochs.

6 Detection Saturates

Table 3 gives pipeline D’s five-fold comparison, the only place in this study where several model families are evaluated under one protocol. A parameter-free range rule (BL-0) that merely checks whether features leave their training min–max attains $F_1 = 0.973$, higher than every learned model including the reference network. Average precision is ≥ 0.98 for all five and 1.000 for three.

Where the separability comes from. Ablating feature groups, shape features alone give $AP = 0.9986$ and amplitude features alone 0.9935, while sensor-relation and current-level features alone give 0.6923 and 0.5052. Removing any single group leaves AP above 0.98. Redundancy across groups of this kind is the signature of a global difference between two recordings, not of a localized fault mechanism.

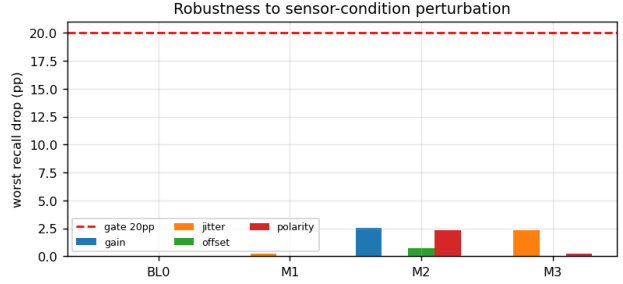


Figure 2: Robustness to benign measurement perturbation. Models that are indistinguishable in Table 3 differ by more than $5\times$.

What does discriminate. We perturbed *normal* held-out data with sensor gain, DC offset, polarity inversion and sampling jitter — changes a maintenance monitor must not report as equipment faults — and measured the increase in false-alarm rate (last column of Table 3, Figure 2). A DC offset of half a training standard deviation drives the Mahalanobis monitor from $\approx 2\%$ to 89.8% of normal windows flagged; the range rule collapses under gain change; PCA-MSPC is most stable at +17.0 points. Recall on fault data is essentially unaffected for all three (≤ 2.4 points), precisely because recall is already saturated — which is why the perturbation test is informative only on the normal side.

A caveat we must state. The shift gate was not evaluated for the LSTM-Autoencoder (marked n/e): the perturbation study requires retraining for every fold $\times$ perturbation combination, and a single five-fold run of that network costs 3.6 h on CPU, exceeding our budget by more than an order of magnitude. Its exclusion from our preferred configuration rests on the false-alarm column alone.

7 Cross-Validation Does Not Predict Deployment

This is the result we did not anticipate and would have missed with a single pipeline.

Pipeline D freezes a deployment configuration: fit on normal blocks 0–2, conformal calibration on block 3, leaving block 4 never seen. On block 4 the frozen system raises a first-stage alarm on $241/2929 = 8.23\%$ of normal windows. The five-fold cross-validation that selected this model reports, *on the identical block 4*, a false-positive rate of $8/2929 = 0.27\%$. The gap is $30\times$.

Nothing about the model differs. The CV fold that holds out block 4 calibrates on block 0; the frozen configuration calibrates on block 3. Block 3 happens to be a quiet stretch, so the quantile it supplies is too tight for block 4, which contains a change in operating regime. Of the 241 non-green windows, 217 fall in original rows 16,000–19,000, and 54 of block 4’s 106 bursts raise at least one alarm.

Table 1: Protocol matrix. “Gap-safe” means windows may not cross an acquisition gap. “Threshold data” is what the decision threshold was fit on. The last column is each pipeline’s own headline F_1 ; §10 explains why these must not be compared.

Pipeline	Model family	Unit	Window	Gap-safe	Threshold data	Own F_1
B	LSTM-AE ablations	window	20	varies by arm	varies by arm	0.684–0.988
C	Isolation Forest	window	10	yes	normal + persistence	0.949
D	PCA-MSPC (+ 4 baselines)	window	10	yes	normal only (conformal)	0.962
E	PCA Q -statistic	row	20	no	normal, locked split	0.958
F	PCA + 2-of-3 alarm	row	20	no	normal, locked split	0.918
G	Logistic regression	burst	native	n/a	supervised CV	0.942

Table 2: Same architecture, different protocol. B arms are 3 seeds; D rows are 3 seeds (G0) and 5 folds (G1). Published row from Ministry of SMEs and Startups and KAIST (2022).

Change from reference		Recall	FPR	F_1
<i>Published</i>				
—	—	0.856	0.0195	0.748
<i>Pipeline B (one factor at a time)</i>				
G0	none (reference)	0.813	0.0254	0.684
A3	whole-window MSE	0.872	0.0108	0.826
A4	normal-only threshold	0.537	0.0000	0.697
P20	all corrections	0.993	0.0019	0.988
<i>Pipeline D (exact reproduction)</i>				
G0	none (reference)	0.843	0.0193	0.743
G1	D’s protocol	1.000	0.0127	0.959

This is a direct, quantified failure of the exchangeability assumption that split-conformal inference requires, arising from blocked rather than random splitting — exactly the splitting that burst structure forces on us. The practical consequence is that **a cross-validated false-alarm rate is not a property of the system**; it is a property of the (fold, calibration block) pair. We know of no way to repair this with the data available: with five blocks and one fault event there is no held-out budget left to estimate the deployment configuration’s false-alarm rate honestly.

Consequence for headline numbers. Recomputing on the frozen system’s own prediction file (held-out normal block plus fault record, prevalence 12.75%) gives first-stage $F_1 = 0.780$ at $\text{FPR} = 0.082$. A two-stage rule that additionally requires a vibration-only monitor to agree for three consecutive windows gives $F_1 = 0.931$, recall 0.876, and $\text{FPR} = 0.0010$ — three false alarms in 2929 windows. The CV average of 0.962 describes neither.

Table 3: One protocol, five model families (pipeline D, five-fold block CV, window unit, prevalence $\approx 12\%$). Detection saturates; the models are ordered only by false alarms and by shift robustness.

	Recall	AP	F_1	FP rate (worst)	Shift (worst)
BL-0 range rule	1.000	1.000	0.973	0.0201	+61.3
M1 Mahalanobis	1.000	1.000	0.965	0.0219	+89.7
M2 Isolation For.	0.932	0.983	0.929	0.0222	+22.3
M3 PCA-MSPC	1.000	0.999	0.962	0.0227	+17.0
BL-1 LSTM-AE (G1)	1.000	1.000	0.959	0.0268	n/e

8 Perfect Ranking, Useless Probabilities

Pipeline D exposes $\text{risk} = 1 - p_{\text{normal}}$, where p_{normal} is a split-conformal p -value. On the frozen evaluation set this score has $\text{AP} \geq 0.99$ — it ranks almost perfectly — yet its Brier score is **0.487** against **0.111** for a constant predictor at the base rate, and its log-loss is 1.695. Of normal held-out windows, 73.1% receive $\text{risk} > 0.5$. A conformal p -value is uniform under the null *by construction*, so $1 - p$ concentrates near 1 for normal data; it is a rank statistic and reporting it as a probability is a category error, however good the ranking.

Realized conformal coverage shows a second, model-dependent failure (Table 4): three monitors track the nominal level within 0.017 at $\alpha = 0.10$, but the range rule saturates at 0.013 regardless of α , because every window inside the training range receives an identical score of zero and the p -value has no resolution below the tie mass. A model whose threshold cannot be read as a probability is unusable for alarm budgeting — and this is invisible in Table 3, where that same model has the best F_1 .

By contrast pipeline E, which fits a sigmoid to its PCA Q -statistic, reports Brier 0.00198 and log-loss 0.0161 on its own locked split. Calibration is therefore achievable on this data — it simply does not come free with a good ranking, and no pipeline that reported only F_1 would have noticed the difference.

Table 4: Realized conformal coverage $\Pr(p \leq \alpha)$ on normal hold-out, averaged over folds.

α	0.01	0.05	0.10
BL-0 range rule	0.008	0.012	0.013
M1 Mahalanobis	0.010	0.047	0.098
M2 Isolation For.	0.011	0.065	0.116
M3 PCA-MSPC	0.012	0.052	0.113

Table 5: Recall vs. available observation length (pipeline G, frozen model, out-of-fold crops). Detectability is bounded by how much contiguous signal a burst provides.

Samples	4	8	10	15	20	30
Recall	0.461	0.467	0.529	0.682	0.749	0.821
F_1	0.631	0.637	0.692	0.811	0.856	0.902

9 Observation Length Bounds Detectability

Pipeline G (supervised logistic regression on five window statistics, 50-fold CV, burst unit) attains recall 0.900 and $F_1 = 0.942$ with zero false positives, and its 20 false negatives fall entirely on two bursts of length 10 and 15. Cropping bursts to a fixed sample count with the model frozen gives the curve in Table 5: recall rises monotonically from 0.461 at four samples to 0.821 at thirty.

This reframes those false negatives. They are not model errors to be tuned away but a statement about information: below roughly 1.5 s of contiguous signal this feature set cannot decide. The operational implication is that a short burst should yield *abstention* rather than a negative classification — a distinction no aggregate F_1 expresses.

10 Why These Numbers Must Not Be Ranked

It is tempting to read the last column of Table 1 as a leaderboard. It is not one, for reasons we think generalise.

The unit differs. D scores windows (prevalence $\approx 12\%$), E and F score rows (prevalence 8.2%), G scores bursts (prevalence 3.4%). F_1 depends on prevalence, so these are different quantities with the same name.

The positive count differs by construction. D’s 428 fault windows, E and F’s 357 positive rows, and G’s fault bursts are different partitions of *one* event. None are independent samples, so no confidence interval computed over them describes uncertainty about faults.

Window validity differs. E’s and F’s main models use 20-row index-contiguous windows that cross acquisition gaps; a 20-row window can span more than 20 s of wall-clock time. Their scores are computed on inputs that pipeline D’s analysis (§3) identifies as artefacts. We report their numbers because they are what those pipelines produced, not because we endorse the inputs.

Selection is entangled across pipelines. E was developed after F’s final evaluation and uses the same test rows, so E’s locked-split guarantee does not survive the cross-pipeline view.

Post-processing changes the operating point, not the model. F’s headline zero false positives comes from a 2-of-3 alarm rule that also reduced true positives from 315 to 303, moving F_1 from 0.936 to 0.918. Reporting the zero without the cost would misstate the trade-off.

We therefore report the spread and refuse the ranking. In our view this is the honest output: a benchmark on which seven protocols yield $F_1 \in [0.684, 0.988]$ with no detection difference is not measuring model quality, and presenting a winner would manufacture a finding the data cannot support.

What can be compared. The refusal is specific to *across* pipelines. Comparison is well defined *within* a protocol, and exactly one of our pipelines supplies one: pipeline D evaluates five model families — including a parameter-free baseline and the reference LSTM-Autoencoder — on identical windows, identical folds, and a calibration set drawn only from normal data (Table 3). That table is the only comparison in this study whose differences are attributable to the models rather than to their evaluation. Its lesson is not that one family wins on F_1 : the parameter-free rule has the highest F_1 and is nonetheless the worst choice, failing both the shift test (§6) and the calibration test (§8). The lesson is that once a single protocol is enforced, the ordering must come from false alarms under shift and from calibration, because detection no longer discriminates. We regard “fix one protocol, then rank on robustness and calibration rather than on F_1 ” as the transferable recommendation of this paper.

11 Threats to Validity

The seven pipelines were built by one group over one dataset; this is a protocol-variation study, not an independent replication, and shared assumptions may have propagated. Several pipelines observed the fault record during exploratory work before their protocols were fixed, so none should be read as a fully held-out evaluation; pipeline D additionally revised three model-selection gates after seeing first results, which it records and which we restate here rather than suppress. Under the original gates the range rule BL-0, not PCA-MSPC, would have been selected. Average precision and AUROC are reported to three decimals in our tables but are not exactly 1 — the largest AUROC we observe is 0.99999880 — and we avoid the phrase “perfect separation” for that reason. Finally, at 10 Hz the Nyquist limit is 5 Hz, far below gear-mesh and bearing fault frequencies (Randall and Antoni, 2011), so we make no spectral-diagnostic claim; this also removes much of the motivation for a high-capacity temporal model.

12 Recommendations

For packaged reference analyses on industrial datasets we recommend reporting, alongside any headline score: (i) acquisition structure, including gap statistics and whether the analysis window fits inside a contiguous segment; (ii) a parameter-free baseline such as a training-range rule, which on this benchmark beats every learned model; (iii) a measurement-shift stress test on *normal* data, which is the only axis that separated models here; (iv) the deployment configuration’s own held-out error, not only a cross-validated average, given the $30\times$ gap we document; and (v) a calibration metric whenever a score is presented as a risk, since ranking and calibration diverge sharply. On this benchmark all five would have changed the conclusion.

13 Conclusion

Seven pipelines, one dataset, F_1 from 0.684 to 0.988, and no detectable difference in detection ability. The spread is produced by windowing, splitting, threshold sourcing, scoring unit and post-processing — choices usually relegated to an implementation appendix. On benchmarks whose labels are confounded with acquisition session, we argue that the useful deliverable is not a better score but an account of what the score is measuring.

The constructive half of that argument is a protocol, and we state it concretely because it is what we would ask a future entrant to adopt: confine windows to contiguous acquisition segments; block cross-validation by burst so that no segment is split; fit every scaler and threshold on normal data from the training blocks alone; include a parameter-free baseline; and report, for the frozen deployment configuration and not only for a cross-validated average, the false-alarm rate under benign measurement perturbation together with a calibration metric. Under that protocol the five model families we tested become genuinely comparable, and the comparison selects a different model from the one F_1 would have selected.

Reproducibility Statement

The pipeline used for §6–§8 runs end to end on CPU from one command with a fixed seed, emits every number in this paper to a named CSV, and is covered by invariant tests asserting that no window crosses an acquisition gap, no burst is split across cross-validation blocks, and no fitted object observes held-out data. Input SHA-256 digests, configuration and environment are recorded in a run manifest. Deleting the output directory and re-running reproduces all tables with zero numerical difference.

References

Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

- Astha Garg, Wenyu Zhang, Jules Samaran, Ramasamy Savitha, and Chuan-Sheng Foo. An evaluation of anomaly detection and diagnosis in multivariate time series. *IEEE Transactions on Neural Networks and Learning Systems*, 33(6):2508–2517, 2022.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.
- J. Edward Jackson and Govind S. Mudholkar. Control procedures for residuals associated with principal component analysis. *Technometrics*, 21(3):341–349, 1979.
- Siwon Kim, Kukjin Choi, Hyun-Soo Choi, Byunghan Lee, and Sungroh Yoon. Towards a rigorous evaluation of time-series anomaly detection. In *AAAI*, 2022.
- Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *ICDM*, 2008.
- Marcos López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.
- Ministry of SMEs and Startups and KAIST. Plastic forming predictive maintenance ai dataset. Korea AI Manufacturing Platform (KAMP), <https://www.kamp-ai.kr/>, December 2022. Smart Manufacturing Innovation Promotion Team (InterX Co., Ltd.).
- S. Joe Qin. Statistical process monitoring: Basics and beyond. *Journal of Chemometrics*, 17(8-9):480–502, 2003.
- Robert B. Randall and Jerome Antoni. Rolling element bearing diagnostics—a tutorial. *Mechanical Systems and Signal Processing*, 25(2):485–520, 2011.
- Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *ICML*, 2019.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- Renjie Wu and Eamonn J. Keogh. Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress. In *IEEE Transactions on Knowledge and Data Engineering*, 2023.
- John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric K. Oermann. Variable generalization performance of a deep learning

model to detect pneumonia in chest radiographs: A cross-sectional study. *PLoS Medicine*, 15(11):e1002683, 2018.